

New Preharvest Treatments and Strategies in Managing *Phytophthora* Brown Rot of Citrus in California

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Abstract

Brown rot caused by *Phytophthora citrophthora*, *P. nicotianae*, *P. syringae*, and *P. hibernalis* is an important fruit disease of citrus in California, and the latter two species are quarantine pathogens in some important export markets. The newly registered fungicides oxathiapiprolin (OXA) and mandipropamid (MAN), as well as a premixture of the two (MAN + OXA) were compared with standard fixed copper and potassium phosphite (KPO₃) treatments (all with different modes of action) under field conditions in two citrus production regions of California. Fruit were sampled periodically over 8 weeks after application in winter or spring seasons, inoculated with zoospores of *P. citrophthora* or *P. syringae*, and brown rot incidence was evaluated. Single applications with all fungicides significantly reduced brown rot incidence of fruit harvested after 8 weeks as compared with the control in seasons with different amounts of precipitation (i.e., 17.2 to 153.9 mm between application and the 8-week sampling). MAN and OXA were similarly or significantly more effective than copper or KPO₃. Two applications done in November and January significantly improved the efficacy of KPO₃

and copper when compared with a single application of each fungicide done in January. For MAN and OXA, however, a single application was similar in efficacy as two applications. Two-application rotations of MAN, OXA, MAN + OXA, or copper significantly reduced the disease incidence by >84% from the control for at least 8 weeks after the second application. Low-volume (935 liters/ha) applications of MAN, OXA, MAN + OXA, or KPO₃, but not copper, were significantly more effective than industry standard high-volume (3,740 liters/ha) applications. Thus, our studies identified and supported registration of new preharvest fungicide treatments to manage brown rot of citrus that are highly effective and persistent, and we optimized treatment strategies. Additionally, rotational programs with fungicides with different modes of action will minimize resistance development in pathogen populations and extend the usage of these fungicides.

Keywords: chemical, crop type, disease management, oomycetes, tree fruits

Phytophthora brown rot, caused by *Phytophthora citrophthora* (R. E. Sm. & R. H. Sm.) Leonian, *P. syringae* (Kleb.) Kleb., *P. nicotianae* Breda de Haan, and *P. hibernalis* Carne, is an important disease of citrus in California (Erwin and Ribeiro 1996; Hao et al. 2018). *P. citrophthora* and *P. syringae* are major causal agents of the disease in the winter harvest season, whereas *P. hibernalis* is considered of minor importance as a brown rot pathogen (Hao et al. 2018). *P. citrophthora* in addition to *P. nicotianae* may also cause losses during warmer environmental conditions (Dirac et al. 2003; Hao et al. 2018; Menge et al. 1988).

Phytophthora brown rot impacts the California citrus industry directly by causing crop losses from pre- and postharvest fruit decay as well as indirectly by export market closures due to quarantine restrictions. Citrus is a major fruit crop in California with a bearing acreage of 109,140 ha and a production value of \$2.3 billion in 2020, and exports amount to hundreds of millions of dollars (U.S. Department of Agriculture 2020). Mature fruit are infected by *Phytophthora* species mostly in the lower tree canopy when rain or irrigation water splashes zoospores and cysts onto the fruit, and infections occur during wet conditions that prevail for at least 3 h. Infected fruit develop brown, leathery lesions and a distinct pungent odor. Lesions remain firm unless secondary infections by other decay pathogens occur. Infected fruit may be symptomless at harvest, but the disease can develop during storage and shipment and spread to surrounding healthy fruit (Eckert and Eaks 1989; Klotz and DeWolfe 1961). Export of California citrus fruit to China, one of the major export markets, was closed from 2013 to 2014 and again in 2015 due to the detection of the quarantine pathogens

P. syringae and *P. hibernalis* in orange fruit shipments. This market closure resulted in economic loss to growers, packinghouses, and shippers (Adaskaveg and Förster 2014).

Disease management strategies implemented to reduce losses from citrus brown rot and detection of quarantine *Phytophthora* spp. in fruit shipments include removal of lower tree branches (i.e., skirting) to diminish the probability of fruit infection, use of proper irrigation practices to minimize canopy wetting, segregation of low-hanging from remaining fruit during harvest, and pre- and postharvest chemical treatments (Adaskaveg et al. 2015; Eckert and Brown 2000; Phillips et al. 1990). For foliar preharvest applications, copper (e.g., copper sulfate in Bordeaux mixtures or fixed coppers such as copper hydroxide and copper oxychloride) and phosphonate (e.g., potassium phosphite, KPO₃) treatments have been used for many years (Adaskaveg et al. 2015). After approximately 40 years of use by the California citrus industry, resistance to KPO₃ was recently detected in isolates of *P. citrophthora*, *P. syringae*, and *P. nicotianae* (Hao et al. 2021). The mode of action of KPO₃ is still unknown; however, several studies suggested that the effect of this compound against *Phytophthora* diseases is a combination of direct inhibition of the pathogens and an induction of plant defense pathways (Cohen and Coffey 1986; Eshraghi et al. 2014; Förster et al. 1998; Guest and Grant 1991; Hao et al. 2021; McGrath 2004; Smillie et al. 1989). Because citrus brown rot caused by resistant isolates of *P. citrophthora* could not be effectively managed by pre- or postharvest treatments with KPO₃, the direct effect against the pathogen appears to be responsible for its efficacy against this disease (Hao et al. 2021).

Mandipropamid (MAN), oxathiapiprolin (OXA), as well as a premixture of these two compounds (MAN + OXA) are Oomycota fungicides that were recently registered or are planned for registration for the management of *Phytophthora* diseases of selected California tree crops. OXA is an oxysterol binding protein inhibitor (Pasteris et al. 2016) with Fungicide Resistance Action Committee (FRAC) Code 49 (FRAC 2022). MAN (FRAC Code 40) is a carboxylic acid amide that targets cellulose synthase in the pathogens, thus, inhibiting cell wall synthesis (Blum et al. 2010). Both are single-site mode of action inhibitors. OXA is considered to have a medium to high risk for resistance development, whereas MAN has a

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low to medium risk (FRAC 2022). To date, natural resistance to each of these two fungicides has not been found. To minimize the risk of resistance developing in pathogen populations, rotational and mixture programs of fungicides with different modes of action are recommended for disease management (Adaskaveg and Förster 2019).

The high in vitro toxicity of MAN and OXA against *Phytophthora* species from citrus was demonstrated previously (Gray et al. 2018), and both fungicides provided excellent efficacy in the management of *Phytophthora* root rot of citrus (Hao et al. 2019). The main objective of the current study was to evaluate their effectiveness as preharvest canopy treatments against *Phytophthora* brown rot in comparison with previously registered fungicides in multiple-year studies at field locations in two major California citrus production areas. Specifically, we evaluated (i) the persistence of preharvest treatments when applied 4 or 8 weeks before harvest; (ii) the efficacy of single- as compared with two-application programs; (iii) the efficacy of mixture/rotation programs that included MAN, OXA, MAN + OXA, KPO₃, and copper; and (iv) treatment efficacy in concentrate (low-volume) as compared with standard dilute (high-volume) foliar spray applications used on citrus.

Materials and Methods

Fungicides and isolates of *Phytophthora* spp. used

Fungicides used in preharvest applications were MAN (Revus; Syngenta Crop Protection, Greensboro, NC, U.S.A.) at 146 g/ha; OXA (Orondis; Syngenta Crop Protection) at 35 g/ha; a mixture of MAN and OXA (MAN + OXA – Orondis Ultra; Syngenta Crop Protection) at 100 g/ha MAN + 12 g/ha OXA; KPO₃ (ProPhyt; Helena Chemical, Collierville, TN, U.S.A.) at 2,351 g/ha; as well as copper (a mixture of copper oxychloride and copper hydroxide – Badge SC or Badge X2; Gowen County, Yuma, AZ, U.S.A.) at 2,227 or 1,975 g/ha of metallic copper/ha for Riverside and Fresno County trials, respectively, that was used in combination with hydrated lime (calcium hydroxide; Fisher Chemical, Fair Lawn, NJ, U.S.A.) at 3,923 g/ha. For postharvest treatments, a premixture of fludioxonil and azoxystrobin (Graduate A+; Syngenta Crop Protection) was applied at 1,200 µg/ml in a canuba based citrus packing coating (Citrus Brite 330; Decco U.S. Post-Harvest, Inc., Monrovia, CA, U.S.A.). *P. citrophthora* isolate 2,440 and *P. syringae* isolate 4,884 obtained from citrus orchards in Ventura and Tulare County, CA, respectively, were used to inoculate orange fruit. Effective concentrations of MAN, OXA, and KPO₃ to inhibit mycelial growth of isolate 2,440 by 50% were 0.0038, 0.0004, and 7.6 µg/ml, respectively, and those for isolate 4,884 were 0.0048, 0.0001, and 13.4 µg/ml, respectively (Gray et al. 2018; Hao et al. 2021). Isolates were cultured on 10% clarified V8 (V8C) agar (Ribeiro 1978) at 20°C, and long-term storage was in liquid nitrogen. Zoospores were produced using the protocol described in Adaskaveg et al. (2015) and adjusted to a concentration of 4×10^4 /ml for inoculation.

Evaluation of the persistence of single preharvest fungicide applications to manage brown rot

Field trials were conducted on 20- to 25-year-old navel orange (*Citrus sinensis* [L.] Osbeck) grafted on “Carrizo” rootstock at the University of California, Riverside (Riverside County) in 2014 to 2017 (four experiments) and on 7- to 12-year-old Thomson Improved – Zimmerman Navel orchard on “Carrizo” rootstock at the Kearney Agricultural Research and Extension Center in Parlier, CA, (Fresno County) in 2014 and 2015 (two experiments). Trees of uniform size were selected, and treatments were arranged in a randomized complete block design in each trial with four single-tree replications for each treatment. MAN, OXA, KPO₃, and copper-lime were applied to the canopy of trees in January or February each year using a backpack air-blast sprayer (SR 430; STIHL Inc., Virginia Beach, VA, U.S.A.) in Riverside County or a tractor-mounted air-blast sprayer (Pak-Blast; Rears Mfg., Coburg, OR, U.S.A.) in Fresno County. Both were calibrated to deliver the equivalent volume of 3,740 liters/ha. Untreated trees were used as controls. Twenty-four fruit were randomly harvested from each tree at least 50 cm above the soil level

(to minimize natural *Phytophthora* spp. infections) 4 and 8 weeks after fungicide application. Each fruit was inoculated with a 30-µl droplet of a zoospore suspension (4×10^4 /ml) of *P. citrophthora* onto the unwounded rind. Inoculated fruit were incubated at 20°C and >95% relative humidity in the dark. Fruit were evaluated for the presence of *Phytophthora* brown rot after 7 to 10 days. A subset of fruit from the 2014 and 2015 Riverside County trials were inoculated with *P. syringae*. Procedures were the same as for *P. citrophthora*, but fruit were precooled to 11°C before inoculation, incubated at this temperature, and evaluated after 4 to 5 weeks. The incidence of disease was calculated based on the number of diseased fruit of the total number of fruit inoculated.

Evaluation of the efficacy of single- compared to two-application programs to manage brown rot

Two field trials were conducted in the Thomson navel orange orchard in Fresno County from November to March 2015/2016 and 2016/2017. Plot layout and applications with MAN, OXA, KPO₃, or a copper-lime mixture were done as described above. Single applications were done in January. For the two-application program, the first treatment was done in November, and the second treatment with the same fungicides was done the following January. Untreated trees were used as controls. Fruit were harvested at least 50 cm above the soil level 2 and 8 weeks after the January application and were inoculated and evaluated for brown rot as described above.

Evaluation of preharvest fungicide rotation/mixture programs to manage brown rot

Field trials were conducted in two navel orange orchards in Riverside and Fresno Counties described above from November to March 2017/2018 and from January to April 2019 (two repeated experiments for each orchard) using the same plot layout and fungicide application method. The first application was done in November 2017 or January 2019 for both orchards and was followed by a second application after 6 to 8 weeks. Treatments included OXA rotated with MAN or copper-lime; MAN rotated with OXA; the MAN + OXA premixture rotated with copper; and MAN + OXA applied in both applications. Treatment application, fruit harvests (4 weeks after the first and 8 weeks after the second applications), inoculation, and disease evaluations were done as described above. Untreated trees were used as controls.

Effect of application volume on the efficacy of preharvest fungicide treatments in managing citrus brown rot

Field trials were conducted in the orchard in Riverside County from March to April 2020 and January to March 2021 (two repeated experiments). The trial layout was a factorial design with application volume and fungicides (four replications each of MAN, OXA, MAN + OXA, KPO₃, and copper-lime) as the two factors in a complete randomized design. Treatments were done using air-blast sprayers calibrated to deliver 3,740 liters/ha (high-volume application) or 935 liters/ha (low-volume application). The same amount of fungicide per hectare was applied using both treatment volumes. Untreated trees were used as controls. Fruit harvested 2, 4, and 8 weeks after application were inoculated and evaluated for brown rot as described above.

Determination of MAN and OXA residues in orange fruit after preharvest application and postharvest processing

Preharvest applications were conducted on ‘Fukumoto’ navel orange grafted on ‘Carrizo’ rootstock in an orchard at the University of California, Riverside (Riverside County), in January 2021 and in the Thomson Improved – Zimmerman Navel orchard in Fresno County described above in November 2017. MAN (146 g/ha) and OXA (17.5 g/ha or 35 g/ha) were applied at 3,740 liters/ha. Plot design was as described above with untreated trees used as controls. Twelve fruit from each of four trees were harvested 1 day after fungicide application. On the same day, half of the fruit were spray-washed for 15 s with a 5% solution of an organic fruit and vegetable cleaner (DeccoNatur Kleen; Decco U.S. Post-Harvest, Inc.) on a brush bed of

an experimental packing line, and this was followed by a 30-s rinse with potable water. Fruit were then treated on a PVC roller bed with fludioxonil-azoxystrobin in a packing carnauba-based fruit coating using a controlled droplet applicator at a rate 1 liter/30,000 kg fruit and allowed to air-dry. Fruit subsamples of approximately 1 kg of preharvest treated, pre-/postharvest treated, and control fruit were submitted to Environmental Micro Analysis Inc. (Woodland, CA, U.S.A.) for MAN and OXA residue analyses using high-performance liquid chromatography tandem mass spectroscopy (HPLC-MS/MS). The limit of detection for both fungicides was 0.01 mg/kg.

Environmental conditions

For all field trial periods, data for daily precipitation and average temperatures was obtained for Stations 44 (UC Riverside, CA; 0.2 km from trial site) and 39 (Parlier, CA; 1 km from trial site) from the California Irrigation Management Information System (CIMIS; <https://cimis.water.ca.gov/Default.aspx>) and are summarized in Table 1. Trial periods with ≤ 45 mm precipitation were considered dry, and those with ≥ 70 mm precipitation were considered wet under California conditions.

Statistical analysis

All statistical analyses were performed in SAS (version 9.4; SAS Institute, Inc., Cary, NC, U.S.A.). Results were considered significant at $P \leq 0.05$. For the single-application preharvest trials, effects of precipitation (dry versus wet trial periods), location (Riverside or Fresno County), fungicide treatment, and harvest time (i.e., fixed effects) on *Phytophthora* brown rot incidence were analyzed using the generalized linear mixed model with the PROC GLIMMIX procedure. Because of significant interactions, effects of treatments on brown rot incidence after inoculation with *P. citrophthora* were analyzed separately for location and harvest time. For the repeated studies on efficacy against *P. syringae*, the effect of fungicide treatment on disease incidence was analyzed.

For the trials on application frequency, the effects of number of applications (one versus two), fungicide treatment, and harvest time (i.e., fixed effects) on brown rot incidence, as well as for trials on efficacy of fungicide rotations/mixtures, the effects of location, harvest time, and fungicide rotation (i.e., fixed effects) on brown rot incidence, were analyzed using the generalized linear mixed model

with the PROC GLIMMIX procedure. Because of significant interactions between treatments and number of applications, as well as between fungicide rotations/mixtures and trial location, fungicide treatment effects on brown rot incidence were analyzed separately for the two trials.

For trials on fungicide application volume, effects of application volume and fungicide (i.e., fixed effects) on brown rot incidence were analyzed using the generalized linear mixed model with the PROC GLIMMIX procedure. Because of a significant interaction between fungicide treatment and volume of application, effects of treatment or sampling time on brown rot incidence were analyzed separately for high- and low-application volumes.

For all GLIMMIX analyses, the response distribution was binomial, and the link function was logit. Trial year was considered a random effect. Least squares means of fungicide treatment or sampling time were computed using the LSMEANS statement with the TUKEY adjustment. Contrast estimates for pairs of treatment were used to determine multiple comparison differences among treatment means.

Results

Weather conditions

Precipitation and ranges of air temperature during the field trials in each year at the two locations are shown in Table 1. During the 2014 and 2015 Riverside and Fresno County trials, ≤ 45 mm rainfall occurred, and these were dry trial periods. During all other trials, there was ≥ 70 mm precipitation, and thus, these were wet trial periods. Average daily temperatures ranged from 5.7 to 25.8°C at the Riverside County and from 2.1 to 20.6°C at the Fresno County sites.

Evaluation of the persistence of single preharvest applications to manage brown rot

The interaction between location (Riverside or Fresno County) and fungicide was significant ($P < 0.0001$); thus, treatment effects were analyzed separately for each location (Fig. 1). The amount of precipitation during the Riverside County trials that were conducted during dry and wet periods, however, did not significantly ($P = 0.0808$) interact with treatment efficacy; therefore, data for all trials were combined. Both studies in Fresno County were conducted

Table 1. Environmental conditions near preharvest field trial locations in Riverside and Fresno County during winter and spring seasons of 2014 to 2021^u

Trial	Location	Year	Number of applications	Precipitation (mm) after fungicide applications				Range of average daily air temperatures (°C)
				To first sampling	First to second sampling	Second to third sampling	Total	
Efficacy of fungicides ^v	Riverside	2014	1	41.0	0.8	NA ^w	41.8	10.3–21.8
		2015	1	25.8	5.8	NA	31.6	10.9–17.3
		2016	1	59.7	15.7	NA	75.4	8.1–24.5
		2017	1	109.0	44.9	NA	153.9	7.4–17.4
	Fresno	2014	1	6.2	33.6	NA	39.8	6.9–16.2
		2015	1	1.3	15.9	NA	17.2	2.1–18.2
Number of fungicide applications ^x	Fresno	2015/2016	1	33.5	77.0	NA	110.5	2.8–16.7
			2	118.1	77.0	NA	195.1	
		2017	1	27.1	60.2	NA	87.3	3.4–20.6
			2	224.4	60.2	NA	284.6	
Rotation of modes of action ^y	Riverside	2017/2018	2	0.8	83.9	NA	84.7	7.5–24.2
		2019	2	93.5	115.6	NA	209.1	6.9–23.7
	Fresno	2017/2018	2	9.5	74.0	NA	83.5	4.2–19.6
		2019	2	43.6	80.5	NA	124.1	4.5–17.5
Volume of application ^z	Riverside	2020	1	75.7	23.9	78.8	178.4	8.7–25.8
		2021	1	41.3	0.3	28.9	70.5	5.7–16.6

^u Daily precipitation and daily average air temperature were obtained for Stations 44 (UC Riverside, CA; 0.2 km from trial site) and 39 (Parlier, CA; 1 km from trial site) from the California Irrigation Management Information System (CIMIS; <https://cimis.water.ca.gov/Default.aspx>).

^v Fruit were sampled 4 and 8 weeks after fungicide applications.

^w Not applicable.

^x Fruit were sampled 2 and 8 weeks after fungicide applications for the one-application trials and after the second application for the two-applications trials.

^y Fruit were sampled 4 weeks after the first and 8 weeks after the second applications.

^z Fruit were sampled 2, 4, and 8 weeks after fungicide applications.

during dry periods. Harvest time (4 or 8 weeks after treatment) had a significant ($P = 0.0005$) effect on fungicide efficacy in the Riverside County but not ($P = 0.9143$) in the Fresno County studies, and data for the two harvests is presented separately. In all trials, fruit from untreated control trees developed a high incidence of brown rot ($\geq 90.8\%$) after inoculation with *P. citrophthora*, and all treatments significantly ($P \leq 0.001$) reduced the incidence from that of the control when applied 4 or 8 weeks before harvest (Fig. 1). In Riverside County, MAN, OXA, and copper were significantly ($P < 0.0001$) more effective than KPO₃, reducing brown rot incidence to between 11.6% (MAN 4 weeks) and 39.1% (MAN 8 weeks), whereas between 81.3 and 84.6% KPO₃-treated fruit developed disease. In contrast, in the Fresno County trials at both harvest dates, MAN, OXA, as well as KPO₃ were significantly ($P \leq 0.0488$) more effective than copper. In the 4-week harvest, the incidence of brown rot was reduced to statistically similar ($P \geq 0.2267$) levels by these three treatments (16.7, 21.7, and 28.3% incidence, respectively), whereas in the 8-week harvest, OXA (26.7% incidence) was significantly ($P < 0.0001$) more effective than KPO₃ (47.5% incidence) but statistically similar ($P = 0.1977$) to MAN (30% incidence).

For fruit from the Riverside County 2014 and 2015 trials that were harvested 4 weeks after application and inoculated with *P. syringae*, all preharvest applications evaluated significantly ($P \leq 0.0003$) reduced the incidence of brown rot from the control, where 49.7% of the fruit developed disease (Table 2). MAN and OXA (both 0.8% incidence) were similarly ($P \geq 0.5318$) highly effective as copper (4.5% incidence), but KPO₃ was significantly ($P \leq 0.0116$) less effective (21.0% incidence).

Evaluation of the efficacy of single- compared to two-application programs to manage brown rot

Both trial seasons when these studies were conducted were considered wet with 195.1 to 284.6 mm accumulated precipitation at the 8-week harvest date. The interaction between number of applications and fungicide treatment on brown rot incidence was significant ($P = 0.0002$); thus, treatment effects were evaluated separately for one- and two-application programs. Additionally, harvest time after application significantly ($P \leq 0.0483$) affected treatment efficacy, justifying the separate presentation of 2- and 8-week harvest data (Fig. 2).

Fruit from untreated control trees developed a brown rot incidence of $\geq 85.8\%$ after inoculation with *P. citrophthora*, and all fungicides significantly ($P \leq 0.0013$) reduced the incidence when fruit were harvested 2 or 8 weeks after treatment (Fig. 2). After one application, MAN and OXA (6.7 and 22.0% incidence and 18.3 and 35%

incidence for the two harvests, respectively) were significantly ($P \leq 0.0430$) more effective than copper (40.7 and 54.2% incidence) at both harvest times. KPO₃ was similar ($P = 0.2189$) in efficacy to MAN, significantly ($P = 0.0209$) less effective than OXA in the 2-week harvest (30.4% incidence), and similar ($P = 0.8899$) to copper in the 8-week harvest (54.2% incidence). After two-applications, all fungicides again significantly ($P < 0.0001$) reduced brown rot incidence from that of the control. The four fungicides were similar ($P \geq 0.0518$) in effectiveness on fruit harvested 2 weeks after treatment, but OXA and KPO₃ were more ($P \leq 0.0196$) effective than copper in the 8-week harvest, and MAN provided intermediate efficacy. The incidence of brown rot on fruit harvested after 8 weeks was 35.5, 20.0, 25.3, and 46.3% for MAN, OXA, KPO₃, and copper, respectively. In pairwise comparisons of the two harvest times for each fungicide, only KPO₃ and copper ($P \leq 0.0287$), but not MAN and OXA ($P \geq 0.8302$), performed significantly better after two as compared with one application.

Evaluation of preharvest fungicide rotation/mixture programs to manage brown rot

The four trials were conducted during wet seasons with 83.5 to 209.1 mm accumulated rainfall at the time of the last harvest (Table 1). There was a significant ($P < 0.0001$) interaction between location (Riverside or Fresno County) and efficacy of fungicide treatments; thus, treatment effects were analyzed separately for each location (Fig. 3). Untreated, inoculated fruit from all harvests

Table 2. Efficacy of single preharvest fungicides applied 4 weeks before harvest to manage brown rot of navel orange fruit inoculated with *Phytophthora syringae*

Treatment (g/ha) ^y	Brown rot incidence (%)
Control	49.7 a ^z
Mandipropamid (146)	0.8 c
Oxathiapiprolin (35)	0.8 c
Potassium phosphite (2,351 g)	21.0 b
Copper (2,227 g metallic copper equivalent)	4.5 c

^y Fungicide applications were done at a research farm in Riverside County in 2014 and 2015, and fruit were harvested after 4 weeks, inoculated with zoospores of *P. syringae*, and incubated at 11°C for 4 to 5 weeks before evaluation.

^z Numbers followed by the same letter do not differ significantly among treatments according to the generalized linear mixed model using PROC GLIMMIX procedure and least square treatment means with the TUKEY adjustment at $P = 0.05$ in SAS.

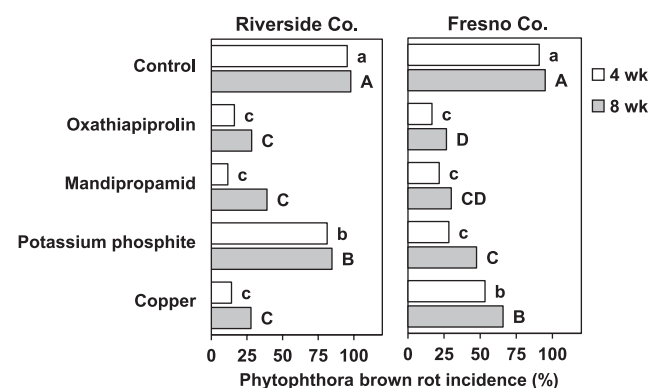


Fig. 1. Efficacy of single preharvest field treatments at 3,740 liters/ha to manage brown rot of navel orange fruit harvested after 4 and 8 weeks and inoculated with *Phytophthora citrophthora*. Trials were done at research farms in Riverside County in 2014 to 2017 and Fresno County in 2014 and 2015. Bars followed by the same lower- or uppercase letter for the 4- or 8-week samplings, respectively, do not differ significantly among treatments according to the generalized linear mixed model using PROC GLIMMIX procedure and least square treatment means with the TUKEY adjustment at $P = 0.05$ in SAS.

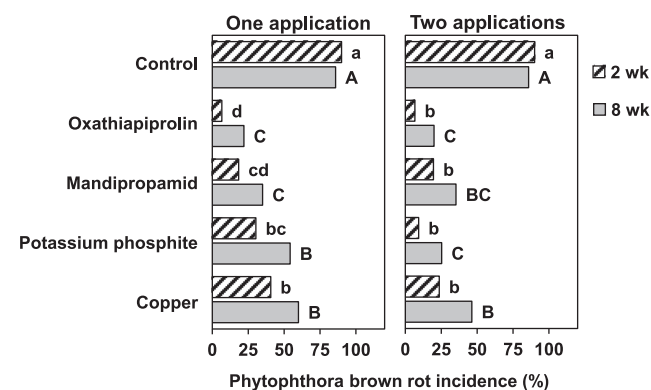


Fig. 2. Efficacy of one- (in January) and two-application (in November and January) preharvest field programs to manage brown rot of navel orange fruit harvested 2 and 8 weeks after the January applications and inoculated with *Phytophthora citrophthora*. Trials were done at a research farm in Fresno County in the winters of 2015/2016 and 2016/2017. Applications were done using a volume of 3,740 liters/ha. Bars followed by the same lower- or uppercase letter for the 2- or 8-week samplings, respectively, do not differ significantly among treatments according to the generalized linear mixed model using PROC GLIMMIX procedure and least square treatment means with the TUKEY adjustment at $P = 0.05$ in SAS.

developed an average incidence of brown rot of >70%, and all treatments significantly ($P < 0.0001$) reduced the incidence 4 weeks after the first application and 8 weeks after the second application as compared with the untreated control. All rotation/mixture fungicide treatments showed similar high efficacy at both sampling times and resulted in an incidence of brown rot of $\leq 9.4\%$ (Fig. 3A) and $\leq 11.5\%$ (Fig 3B) in Riverside and Fresno County trials, respectively.

Effect of application volume on the efficacy of preharvest fungicide treatments in managing citrus brown rot

Studies were done in Riverside County during two seasons with high precipitation (Table 1). All treatments at all timings again significantly ($P \leq 0.0455$) reduced the incidence of brown rot from the controls where 77.1 to 93.1% of the fruit developed decay. Based on a factorial analysis, there was a significant ($P < 0.0001$) interaction between fungicide performance and application volume; thus, data is presented separately for the two application methods (Fig. 4). MAN and the MAN + OXA premixture were significantly ($P \leq 0.0018$) more effective using an application volume of 935 liters/ha as compared with 3,740 liters/ha when fruit were harvested and inoculated 8 weeks but not 2 or 4 weeks after treatment. For KPO₃, the incidence of brown rot was significantly ($P < 0.0001$) lower at all three harvest timings when applied at a low volume. In contrast, OXA and copper showed a similar ($P \geq 0.84$) high effectiveness using both application volumes at all sampling times. Overall, when applied at 935 liters/ha, the incidence of brown rot was reduced to between 0.5 and 18.9% by the treatments evaluated for the three harvest intervals. The five fungicides performed statistically similar

($P \geq 0.1359$) in the 4-week after application harvest, whereas in the 2-week harvest, KPO₃ was significantly ($P \leq 0.0062$) less effective as compared with all other treatments, and in the 8-week harvest, KPO₃ was significantly ($P \leq 0.0464$) less effective than MAN, OXA, and MAN + OXA.

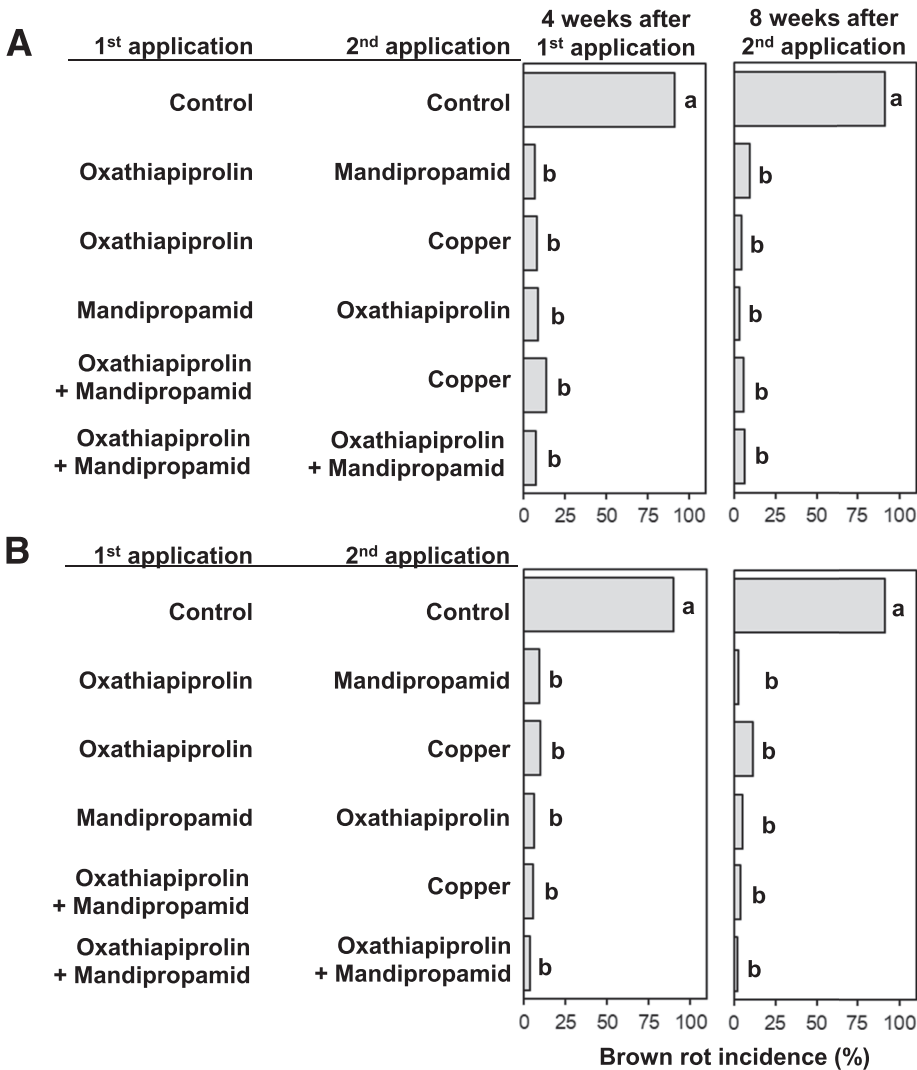
MAN and OXA residues in orange fruit after preharvest application and postharvest processing

MAN and OXA were not detected by HPLC-MS/MS analyses in untreated control fruit (Table 3). Residue levels in orange fruit pre-harvest treated with 146 g/ha MAN or with 17.5 or 35 g/ha OXA were 0.2, 0.025, or 0.01 mg/kg, respectively. After a postharvest wash with a commercial fruit cleaner on a brush bed of an experimental packing line that was followed by a water rinse, both fungicides were no longer detectable at a limit of detection of 0.01 mg/kg.

Discussion

Field trials in this study that were conducted in two major citrus growing areas of California demonstrated the high and consistent efficacy of protective preharvest applications with MAN and OXA in managing *Phytophthora* brown rot caused by *P. citrophthora* or *P. syringae* that was similar to or higher in efficacy than for copper or KPO₃. Although different amounts of precipitation occurred during the trial periods over 8 years, results for the two new fungicides, as well as the MAN + OXA premixture, were mostly very similar among years and between the two field sites. Trials, however, also need be conducted in other citrus growing regions that experience

Fig. 3. Efficacy of preharvest fungicide rotation field programs to manage brown rot of navel orange fruit harvested 4 weeks after the first and 8 weeks after the second application and inoculated with *Phytophthora citrophthora*. Trials were done at research farms in **A**, Riverside and **B**, Fresno County in the winters of 2017/2018 and 2019. Fungicides were applied at 3,740 liters/ha in November 2017 and January 2018 or in January and March 2019. Bars followed by the same letter do not differ significantly according to the generalized linear mixed model using PROC GLIMMIX procedure and least square treatment means with the TUKEY adjustment at $P = 0.05$ in SAS.



much higher rainfall and warmer temperatures during the fruit ripening stages such as Florida. MAN and OXA were found to persist for a minimum of 8 weeks after application under our field conditions, and moreover, a two-application program did not increase efficacy as compared with a single application. This contrasted with standard treatments with KPO₃ or copper where two applications resulted in significantly less brown rot than a single application when fruit were harvested 8 weeks after the last application. Additionally, copper was significantly less effective in Fresno County, whereas KPO₃ was less effective in Riverside County. Different formulations of a copper hydroxide/copper oxychloride product and slightly different amounts of metallic copper equivalent were used at the two locations (i.e., a liquid soluble concentrate at 2,227 g/ha in the Riverside County trials and a dry formulation at 1,975 g/ha in the Fresno County trials). This possibly can explain the higher efficacy in the Riverside County trials. Still, the reason for the difference in KPO₃ efficacy between the two sites is not known. Trials in Riverside County were conducted during dry and wet periods; thus, the amount of rainfall did not significantly affect the performance of

KPO₃. Perhaps tree size and age, as well as canopy volume may have influenced the results.

Preharvest applications at a low volume (935 liters/ha) to the 24- to 25-year-old orange trees used in our trials with all fungicides, especially KPO₃, were as effective or more effective in reducing brown rot as compared with the standard high-volume (3,740 liters/ha) application. The higher persistence of low-volume applications was evident by the improved decay control in the 8-week harvest and also in the 2- and 4-week harvests for KPO₃. This is an important finding because the lower spray volume not only reduces water usage but also allows for more acreage to be treated with a single tank load, resulting in time (i.e., fewer tank refills) and cost savings. High-volume sprays are being routinely used by the California citrus industry for many years to provide coverage of the high density of the citrus tree canopy (J. E. Adaskaveg, *personal communications* with representatives of the California citrus industry). In a study published in 1972, however, both high- and low-volume sprays with copper provided excellent protection of citrus against brown rot (Klotz et al. 1972). Modern air-blast sprayer technology likely contributes to the high efficacy of low-volume applications to citrus. Still, for older citrus plantings, higher volumes may need to be used. Ideally, spray volume should be calculated using the tree row volume method that is based on the estimated volume of foliage and is related to tree age and planting density (Scapin et al. 2015). Using this method, spray volumes have been successfully reduced in the management of citrus canker with copper (Scapin et al. 2015) and for disease management of apple (Sutton and Unrath 1984). This method, however, has not been adopted by California citrus growers. In a study with 44-year-old mandarin trees, spray volumes above 2,000 liters/ha resulted in increased canopy run-off of the applied treatment (Cunningham and Harden 1998).

MAN and OXA, as well as MAN + OXA, were also effectively integrated into rotation programs with copper, and brown rot incidence was reduced to very low levels. This enables the implementation of resistance management programs using fungicides with different modes of action when under highly favorable conditions two applications need to be applied before harvest. KPO₃ was not used in these preharvest rotation trials because it is the only effective fungicide registered for postharvest use to manage brown rot (Adaskaveg and Förster 2014; Adaskaveg et al. 2015), and ideally, in a resistance management program, the same mode of action should not be applied in the field and to harvested fruit in the packinghouse (Adaskaveg and Förster 2019, 2022). Resistance to MAN or OXA has not been detected to date in citrus pathogens; however, resistance to KPO₃ was found in recent surveys of brown rot pathogens including *P. citrophthora*, *P. syringae*, and *P. nicotianae* (Hao et al. 2021). In previous research, brown rot caused by KPO₃-resistant isolates of *P. citrophthora* was effectively managed using preharvest applications with OXA (MAN was not evaluated in this research), whereas KPO₃ was not effective (Hao et al. 2021). Overall, in our current studies, efficacy of KPO₃ was variable and was highest when applied at low volume and using two applications. The high postharvest efficacy of KPO₃ is based on its outstanding postinfection activity where treatments applied 18 to 24 h after inoculation with zoospores of *Phytophthora* sp. still reduce brown rot incidence to very low levels (Adaskaveg et al. 2015). MAN (Adaskaveg et al. 2013) and OXA (Adaskaveg *unpublished*) performed very poorly in similar postinoculation studies; therefore, these two fungicides are strictly protective treatments that need to be applied before infection events.

Studies in this research were all done on navel orange. This is because navel oranges are harvested in the winter season when most of the rainfall and *Phytophthora* spp. infection periods occur in California. Furthermore, the quarantine pathogens *P. syringae* and *P. hibernalis* only cause disease during the cooler winter seasons (Hao et al. 2018), and they were detected on brown-rotted navel orange fruit in export markets. In other studies (Adaskaveg, *unpublished*), high efficacy of MAN and OXA was also demonstrated on lemon and mandarin. In most of our trials, we used the non-quarantine pathogen *P. citrophthora* for fruit inoculation. Inoculations with the quarantine pathogen *P. syringae* that is currently the other main causal agent of citrus brown rot in California are more difficult to conduct. In

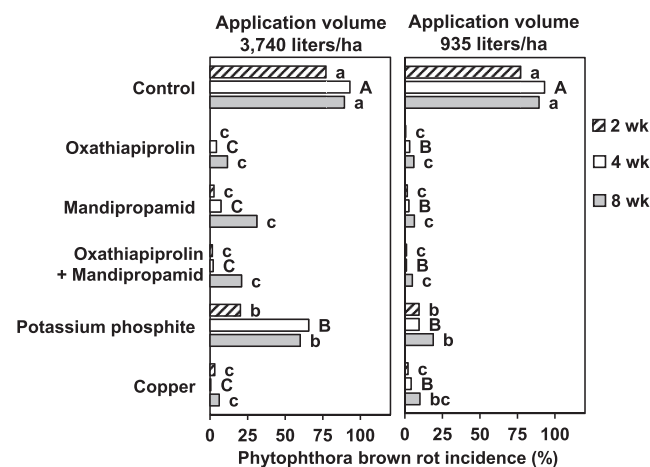


Fig. 4. Efficacy of preharvest field treatments applied at high (3,740 liters/ha) or low (935 liters/ha) volumes to manage brown rot of navel orange fruit harvested after 2, 4, and 8 weeks and inoculated with *Phytophthora citrophthora*. Applications were done in March 2020 or January 2021 in Riverside County. Bars followed by the same lower- or uppercase letter do not differ significantly among treatments for each sampling time according to the generalized linear mixed model using PROC GLIMMIX procedure and least square treatment means with the TUKEY adjustment at $P = 0.05$ in SAS.

Table 3. Fungicide residues on pre- and postharvest treated navel orange fruit

Preharvest treatment (application rate g/ha) ^w	Postharvest treatment ^x	Fungicide residue amount (mg/kg ± SE) ^y
Untreated control	—	ND ^z
Oxathiapiprolin (17.5)	—	0.025 ± 0.015
Oxathiapiprolin (35)	—	0.01 ± 0
Oxathiapiprolin (17.5)	Packing line	ND
Oxathiapiprolin (35)	Packing line	ND
Untreated control	—	ND
Mandipropamid (146)	—	0.2 ± 0
Mandipropamid (146)	Packing line	ND

^w Preharvest treatments of oxathiapiprolin and mandipropamid were applied to the canopy of trees using backpack air-blast sprayers at 3,740 liters/ha.

^x Fruit were harvested 1 day after fungicide application and immediately processed on an experimental packing line including washes with a cleaner, a rinse with potable water, and a fungicide application in a carnauba-based citrus coating.

^y The limits of detection for high-performance liquid chromatography tandem mass spectroscopy analyses were 0.01 mg/kg. Residue values shown are the mean of two field trials ± standard error (SE).

^z ND = nondetectable residue level.

our and other studies with *P. syringae*, fruit needed to be incubated at 5 to 12°C for extended periods of time (≥ 4 weeks), and decay development was not as consistent as for *P. citrophthora* (Adaskaveg et al. 2015). As for *P. hibernalis*, it is currently rarely found to cause fruit brown rot (Hao et al. 2018). Furthermore, treatment efficacy in our trials was based on inoculation of harvested fruit. Although brown rot was present in our experimental orchards, it was not uniformly distributed among trees; thus, a larger number of trees would have to be used for each treatment replication to obtain statistically significant results. Additionally, brown rot mostly occurs after periods of rainfall, and this would have limited the number of experiments that could have been conducted.

Based on our research, OXA was registered on citrus for both foliar and soil applications against *Phytophthora* diseases in 2019, the label for MAN was expanded from soil application for citrus nursery plants to also include foliar applications of mature trees in the field in 2020, and the premixture MAN + OXA was registered as a foliar application for citrus also in 2020. The registration of MAN and OXA will help minimize overall crop losses from brown rot, facilitate the implementation of resistance management strategies, and reduce the citrus industry's dependency on copper that has become under regulatory scrutiny due to concerns about soil and watershed contamination after many decades of usage. Development of rotation programs with older (i.e., KPO_3 , copper products) and new modes of action will provide a sustainable management strategy for *Phytophthora* diseases of citrus and extend the usage of all registrations. The new preharvest treatments will help prevent field infections that could develop into brown rot during transit on nonprotected fruit and be detected at ports of arrival in export counties. Maximum residue limits for MAN and OXA are currently being established in these destinations. Still, our fungicide residue analyses indicate that postharvest processing (i.e., washing with a fruit cleaner, followed by a water rinse on a brush bed and postharvest fungicide treatment) of fruit preharvest treated with MAN or OXA reduced residues below the level of detection.

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